

Requirements Summary

Objective

Build an **Autonomous Trading Agent** that:

1. Reads signals from CoinMarketCap (CMC Agent Hub).
2. Makes trading decisions automatically.
3. Signs transactions through Trust Wallet Agent Kit (TWAK).
4. Executes trades on BNB Chain.
5. Uses PancakeSwap and/or BSC Perpetuals.
6. Operates according to user-defined trading rules.
7. Runs autonomously with minimal human intervention.

Recommended Stack

Layer 1 – Data & Signals

CoinMarketCap Agent Hub

Provides:

- Market Data API
- Data MCP
- Technical Indicators
- News
- Sentiment Analysis
- On-chain Data
- Wallet Data
- KOL/Social Signals
- Skills Marketplace

Layer 2 – Custody & Execution

Trust Wallet Agent Kit (TWAK)

Provides:

- Wallet Management
- Local Signing
- Secure Transaction Approval
- Autonomous Execution

Layer 3 – Chain & Trading

BNB AI Agent SDK

Provides:

- BSC Mainnet Integration
 - PancakeSwap Trading
 - BSC Perpetual Futures
 - Agent Execution Primitives
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Project Name

AlphaTrader AI

An Autonomous Multi-Agent Trading Intelligence Platform

Instead of building a simple buy/sell bot, build a system of specialized AI agents.

This demonstrates:

- Technical complexity
 - Real-world utility
 - Advanced AI architecture
 - Maximum use of sponsor technologies
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SRS

AlphaTrader AI

Autonomous Multi-Agent Trading Platform

1. Executive Summary

AlphaTrader AI is a fully autonomous cryptocurrency trading platform that combines CoinMarketCap intelligence, AI reasoning, Trust Wallet execution, and BNB Chain trading infrastructure.

The platform continuously monitors:

- Market trends
- Technical indicators
- Social sentiment
- News events
- On-chain activities

and executes trades automatically while following risk controls and user-defined strategies.

2. Business Goals

Primary Goal

Create an AI trader capable of:

- Analyzing market conditions
- Making independent decisions
- Executing trades
- Managing risk
- Optimizing returns

Secondary Goals

- Reduce emotional trading
 - Provide 24/7 trading
 - Improve trade execution speed
 - Minimize user intervention
-

3. User Roles

Trader

Creates:

- Trading profiles
- Risk settings
- Capital allocation

Monitors:

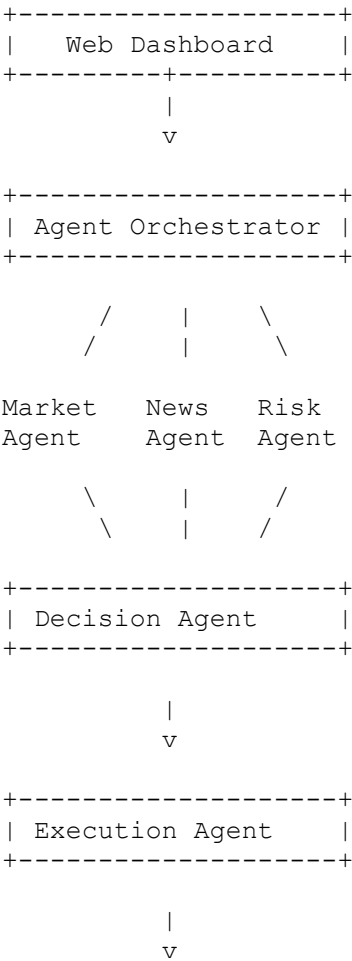
- Performance
- Open positions

Admin

Manages:

- Users
- Agents
- API Keys
- System Health

4. System Architecture



Trust Wallet Agent Kit

|
v

BNB Chain

|
v

PancakeSwap

5. AI Agent Modules

Agent 1: Market Intelligence Agent

Purpose

Analyze market conditions.

Data Sources

CMC APIs

Responsibilities

Monitor:

- Price movement
- Volume
- Liquidity
- Volatility

Generate:

- Buy signals
 - Sell signals
-

Agent 2: Technical Analysis Agent

Purpose

Generate technical indicators.

Indicators

RSI

MACD

EMA

SMA

Bollinger Bands

VWAP

ATR

Fibonacci

As per strategies defined in section 8

Outputs

Bullish Score

Bearish Score

Trend Strength

Agent 3: News & Sentiment Agent

Purpose

Monitor market sentiment.

Sources

CMC News

CMC MCP

Social Signals

KOL Tracking

Analyze

Positive sentiment

Negative sentiment

Breaking events

Output

Sentiment Score

0-100

Agent 4: Whale Monitoring Agent

Purpose

Detect smart money activity.

Monitor

Large transactions

Exchange inflows

Exchange outflows

Wallet accumulation

Wallet dumping

Output

Whale Confidence Score

Agent 5: Risk Management Agent

Purpose

Protect capital.

Controls

Maximum daily loss

Maximum drawdown

Position size limits

Trade frequency limits

Leverage limits

Example

Daily loss exceeds 5%

→ Suspend trading

Agent 6: Decision Agent

Purpose

Combine outputs from all agents.

Inputs

Market Score

Technical Score

Sentiment Score

Whale Score

Risk Score

AI Decision

BUY

SELL

HOLD

SHORT

CLOSE

Agent 7: Execution Agent

Purpose

Execute trades.

Actions

Swap Token

Open Position

Close Position

Set Stop Loss

Set Take Profit

Integrations

TWAK

BNB AI SDK

PancakeSwap

6. Functional Requirements

FR-001 User Registration

Users shall create accounts.

FR-002 Wallet Connection

Support:

- Trust Wallet
 - WalletConnect
-

FR-003 Capital Allocation

User defines:

- Trading Capital
 - Max Exposure
-

FR-004 Strategy Configuration

User defines:

Conservative

Balanced

Aggressive

FR-005 Autonomous Trading

Agent shall trade automatically.

FR-006 Live Trade Monitoring

Display:

- Entry Price
 - Current Price
 - P&L
-

FR-007 Risk Controls

Stop trading when risk limits are breached.

FR-008 Trading Logs

Store:

- Signals
 - Decisions
 - Executions
-

FR-009 Explainable AI

Every trade shall include:

Why Trade Happened

Example:

BTC RSI oversold, positive sentiment, whale accumulation detected.

FR-010 Backtesting

Simulate strategy performance.

7. Dashboard Pages

Login

Home Dashboard

Display:

- Portfolio Value
 - Daily P&L
 - Win Rate
-

Agent Center

Show:

- Agent Status
 - Last Decision
 - Confidence
-

Live Trades

Open positions

Closed positions

Market Intelligence

Trending Tokens

Sentiment Heatmap

Whale Tracker

Risk Dashboard

Drawdown

Exposure

Leverage

Settings

Trading Rules

Risk Rules

Notifications

8. Recommended Strategy

Multi-Factor AI Momentum Strategy

The agent enters trades only when several factors align.

Signal Sources

Factor	Weight
Technical Analysis	35%
Market Sentiment	25%
Volume & Momentum	20%
Whale Activity	10%
Risk Assessment	10%

Buy Conditions

Technical Score > 70

AND

Sentiment Score > 60

AND

Volume Increase > 30%

AND

Whale Accumulation = True

AND

Market Trend = Bullish

Sell Conditions

Technical Score < 40

OR

Negative Sentiment Spike

OR

Whale Distribution

OR

Stop Loss Triggered

Strategy 1: Smart Money Tracker (My Favorite)

This would be very attractive to judges.

Logic

Monitor:

- Whale wallets
- Large transactions
- Exchange inflows
- Exchange outflows

Example

Whales purchase \$5M of a token.

Exchange reserves decrease.

Sentiment becomes positive.

Technical breakout occurs.

Agent buys automatically.

Why It Wins

Uses:

- CMC Data
- On-chain signals
- AI reasoning

Much more sophisticated than RSI bots.

Strategy 2: Narrative Rotation Strategy

This is how many crypto funds trade.

AI Detects Narratives

Examples:

- AI Coins
- Meme Coins
- RWA
- DeFi
- Gaming
- Layer 2

Example

AI sector:

- FET
- AGIX
- OCEAN

all begin trending.

News sentiment rises.

Volume spikes.

Agent rotates capital into AI narrative.

Exit

Narrative loses momentum.

Agent exits positions.

Strategy 3: News Event Trading

Monitor

CMC News

Social Mentions

Breaking Events

Example

BNB announces major partnership.

Sentiment becomes strongly positive.

Volume surges.

Agent buys BNB.

Exit

Momentum weakens.

Strategy 4: Fear & Greed Strategy

Contrarian strategy.

Buy

Fear & Greed < 25

Market oversold.

Sell

Fear & Greed > 80

Market euphoric.

This works surprisingly well in crypto.

Strategy 5: PancakeSwap New Trend Hunter

Because the hackathon is BNB-focused.

Agent Monitors

New BSC tokens.

Volume growth.

Liquidity increases.

Holder growth.

Social mentions.

Buy

Liquidity > \$100k

Volume growth > 50%

Holder growth > 20%

Positive sentiment

Sell

Volume drops.

Whales exit.

Sentiment turns negative.

Strategy 6: AI Consensus Strategy (Best for Winning)

Instead of one strategy, create five specialized agents.

Agent A

Technical Analysis

Agent B

News Analysis

Agent C

Sentiment Analysis

Agent D

Whale Tracking

Agent E

Risk Assessment

Each agent votes.

Example:

Agent	Vote
Technical	Buy
Sentiment	Buy
Whale	Buy
News	Buy
Risk	Hold

Result:

4/5 Buy

Decision Agent executes trade.

9. Non-Functional Requirements

Performance

Decision latency

< 2 seconds

Trade execution

< 5 seconds

Security

AES-256

TLS 1.3

Role-Based Access

Availability

99.9%

Scalability

100,000 users

10,000 concurrent agents

10. CoinMarketCap Integration

Market Data API

Real-time prices

Data MCP

Technical indicators

Skills Marketplace

Trading skills

x402

Pay-per-call services

Agent Hub

Agent orchestration

11. Trust Wallet Integration

TWAK

- Wallet Creation
- Signing
- Trade Authorization
- Autonomous Execution

12. BNB Chain Integration

BSC Mainnet

Execute swaps

PancakeSwap

Spot trading

BSC Perpetuals

Leverage trading

Yes, but I would **not limit AlphaTrader AI to a simple "Select Trading Pair" dropdown.**

For Track 1, judges will likely be more impressed if users can choose **how much control they want to give the agent.**

Recommended Design

Agent Creation Wizard

Instead of:

✕ Select Pair → Start Bot

Use:

✓ Create Trading Agent

Step 1: Agent Profile

Agent Name

- FIDA Hunter
- Meme Coin Scout
- AI Sector Trader

Description

- Optional notes
-

Step 2: Strategy Selection

Select one or more strategies:

- ☐ AI Consensus Strategy
- ☐ Daily Engulfing Reversal
- ☐ Narrative Rotation

☐ Whale Tracking

☐ Fear & Greed Contrarian

☐ Custom Strategy

Step 3: Market Universe

Option A: Single Pair

- FIDA/USDT
- BNB/USDT
- BTC/USDT

Option B: Multiple Pairs

Select up to 10 pairs

Option C: Agent Decides ★

Examples:

- AI Tokens
- Meme Coins
- DeFi Tokens
- Top 100 Coins
- BNB Ecosystem

The agent scans and finds opportunities.

This is much more "autonomous".

Step 4: Risk Management

Maximum risk per trade:

- 1%
- 2%
- 5%

Maximum daily loss:

- 5%
- 10%

Maximum open positions:

- 1
- 3
- 5

Step 5: Entry Filters

User can enable:

- ☒ RSI Confirmation
- ☒ Volume Confirmation
- ☒ Whale Confirmation
- ☒ Sentiment Confirmation
- ☒ Fear & Greed Confirmation

Step 6: Execution Rules

Trade Type:

- ☐ Spot
- ☐ Perpetual
- ☐ Both

Execution:

- ☐ Fully Autonomous

☐ Ask Approval

☐ Paper Trading

Step 7: Natural Language Strategy Builder ★

This aligns perfectly with the hackathon requirements.

User enters:

Trade AI coins using Daily Engulfing and Whale Accumulation. Risk 2% per trade. Use 1:3 risk reward. Stop trading after 3 losses.

AlphaTrader AI converts it into:

Strategy:
DailyEngulfing

Assets:
AI Tokens

Risk:
2%

RR:
1:3

Filters:
Whale Accumulation
Volume Spike

Max Consecutive Losses:
3

This is exactly:

Natural-language strategy in → on-chain execution out

which is explicitly mentioned in Track 1.

Agent Dashboard

Each agent should display:

Agent Status

☐ Running

Selected Assets

- FIDA
- BNB
- CAKE

Strategy

Daily Engulfing + Whale Tracking

Confidence Score

87%

Open Positions

3

Today's PnL

+\$182

Last Decision

BUY FIDA

Reason:

- Bullish engulfing
- RSI > 50
- Volume 180% above average
- Whale accumulation detected

Trading Mode

Manual Pair Mode

User selects:

- FIDA/USDT
- BNB/USDT

Good for beginners.

Category Mode ★

User selects:

- AI Tokens
- Meme Coins
- DeFi

Agent finds opportunities.

Autonomous Discovery Mode ★ ★ ★

User says:

Find the best trading opportunities on BNB Chain using AI Consensus Strategy.

Agent:

1. Scans hundreds of tokens.
2. Scores them.
3. Chooses the best setup.
4. Executes trades automatically.

This third mode best demonstrates an **Autonomous Trading Agent**, which is exactly what Track 1 is designed to reward.